

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 4+ years of experience building production-grade TypeScript/Node.js systems on Google Cloud Platform. Led zero-downtime migration of 20+ applications from Parse/MongoDB to Firebase/GCP, implementing real-time data pipelines with Pub/Sub and saving \$5K/month in infrastructure costs. Strong system architecture background including microservices with Apache Kafka and Kubernetes. Proven track record of remote collaboration across US and European timezones with high technical ownership and reliability-focused engineering.

CORE COMPETENCIES

TypeScript/Node.js Backend

Google Cloud Platform

System Architecture

Zero-Downtime Migrations

Real-time Data Pipelines

PostgreSQL & NoSQL

API Design

Remote Collaboration

WORK EXPERIENCE

MTailor

September 2022 – Present

Software Engineer (US Remote)

- Led **zero-downtime migration** of 20+ applications from Parse/MongoDB to Firebase/GCP, reporting directly to CTO
- Built **real-time two-way sync** between MongoDB and Firestore using Node.js and Google Pub/Sub for reliable message processing
- Developed 3D visualization feature using video overlay and ffmpeg, increasing buyer conversion
- Migrated file storage from Amazon S3 to GCS via Python script, ensuring data integrity
- Implemented Express Shipping feature, generating **\$40 additional revenue per order**
- Saved company **\$5,000/month** by consolidating all services on Google Cloud Platform

MTailor

March 2022 – July 2022

Contractor (US Remote)

- Built TypeScript migration script to read MongoDB data and write to Firestore with complex processing logic
- Authored technical documentation for Firebase SDKs (Node.js and Python) for engineering team
- Trained Operations team on Firebase Dashboard and backup procedures

Dr Wealth

August 2021 – January 2022

Software Engineer (Singapore Remote)

- Led optimization of TypeScript PWA delivering **real-time stock prices** from Firebase backend
- Built data pipelines querying **2+ million Firestore records** to sync prices from Morningstar APIs
- Extended backend APIs using Firebase Cloud Functions and Express.js
- Added responsive UI components using React and Tailwind CSS

Mind2matter.co

January 2022 – July 2022

Software Engineer (US Remote)

- Built TypeScript backends for DeFi applications using Web3 and Node.js
- Delivered React UIs from Figma designs under tight timelines

CodeBits

January 2020 – July 2021

Software Engineer (Uganda Remote)

- Led team of 4 developers building legal tech systems for NGOs in Uganda
- Architected microservices backend** for FIDA Uganda case management using Apache Kafka, Docker, and Kubernetes
- Built cross-platform mobile app for LASPNET using React Native and Expo with Firebase push notifications
- Developed USSD service for legal aid providers communicating with Node.js backend via gRPC

PROJECTS

Parse → Firebase Migration Platform GCP

End-to-end migration system with real-time Pub/Sub sync, zero-downtime cutover, and automated data validation

Node.js, TypeScript, Firebase, Google Pub/Sub, MongoDB, Python

Real-time Stock Price Pipeline Firebase

High-volume data pipeline syncing 2M+ records from Morningstar APIs to Firestore

TypeScript, Firebase Cloud Functions, Express.js, REST APIs

Legal Case Management System Microservices

Microservices architecture for NGO case management with mobile app and USSD integration

Node.js, Apache Kafka, Docker, Kubernetes, React Native, gRPC

EDUCATION

Bachelor of Science in Computer Engineering — [Makerere University](#)

2018 – 2022

Kampala, Uganda

SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Go (intermediate)

Backend: Node.js, Express.js, Firebase Cloud Functions, REST APIs, GraphQL (learning) **Databases:** PostgreSQL, MongoDB, Firestore, Redis

Cloud: Google Cloud Platform (Pub/Sub, GCS, Cloud Functions), AWS (S3, EC2), Heroku

Infrastructure: Docker, Kubernetes, Apache Kafka, CI/CD pipelines **Frontend:** React, React Native, Tailwind CSS, HTML/CSS